Math 421

Monday, October 27

Announcements

- Grace’s Drop-In Hours
  - Wednesday 12-1 pm VV 311
  - Thursday 9-11 am VV B205
- MLC
  - Proof Table: M-Th 3-7:30 VV B227
  - Course Assistant: Th 4-7 Table 4
- Homework 7 Due Friday

Activity

Give sequences that exhibit the following properties:

- The set $\{n : s_n > \limsup s_n\}$ can be infinite.
- The set $\{n : s_n < \liminf s_n\}$ can be infinite.
- If $\liminf s_n < \limsup s_n$, the set

$$\{n : \liminf s_n \leq s_n \leq \limsup s_n\}$$

can be empty. **Hint:** Try to sketch out what this property is saying, what would need to be true about a sequence that satisfies this property?

$\liminf$, $\limsup$ and the set of subsequential limits

Theorem 11.7. Let (s_n) be any sequence. There exists a subsequence whose limit is $\limsup s_n$, and there exists a subsequence whose limit is $\liminf s_n$.

Theorem 11.8. Let (s_n) be any sequence in $\mathbb{R}$, and let S denote the set of subsequential limits of (s_n).

1. S is nonempty.
2. $\sup S = \limsup s_n$ and $\inf S = \liminf s_n$.
3. $\lim s_n$ exists if and only if S has exactly one element, namely $\lim s_n$.

Examples

1. (s_n) where $s_n = \frac{1}{n}$. $S = \{0\}$

2. (a_n) where $a_n = n^2(-1)^n$. $A = \{-\infty, +\infty\}$.

3. (b_n) where $b_n = \sin\left(\frac{n\pi}{3}\right)$. $B = \left\{-\frac{\sqrt{3}}{2}, 0, \frac{\sqrt{3}}{2}\right\}$.

4. (r_n) a list of all rational numbers. $R = \mathbb{R} \cup \{-\infty, +\infty\}$.

5. (c_n) where $c_n = n(1 + (-1)^n)$. $C = \{0, +\infty\}$.

Chapter 2: Sequences

Section 10 (Part 3): Cauchy Sequences

Cauchy Sequences

Def: A sequence (s_n) of real numbers is called a *Cauchy sequence* if for each $\epsilon > 0$ there exists a number N such that

Q: What is this definition saying in your own words? How does this definition compare to the definition of a convergent sequence?

Convergent ⇒ Cauchy

Lemma 10.9. Convergent sequences are Cauchy sequences.

Activity: Sketch a proof of the Lemma.

➊ What are you given?

➋ What are you trying to show?

➌ Sketch the details.

Cauchy ⇒ Bounded

Lemma 10.10. Cauchy sequences are bounded.

Proof.

Convergent $\Leftrightarrow$ Cauchy

Theorem 10.11. A sequence is a convergent sequence if and only if it is a Cauchy sequence.

Q: Why might this theorem be useful?

Cauchy $\Rightarrow$ Convergent

Discussion.

❶ What are we given?

❷ What are we trying to show?

❸ What theorems/properties/definitions might be useful?

Recall – From Ross §14

- Consider the series given by $\sum_{n=1}^{\infty} \frac{n}{n^2+3}$. Find the first 4 terms in its sequence of partial sums.

- What does it mean for an infinite series to converge?

- What does it mean for an infinite series to be absolutely convergent?

Cauchy Criterion

Def. We say a series $\sum a_n$ satisfies the Cauchy criterion if its sequence of partial sums is a Cauchy sequence.

Theorem 14.4. A series converges if and only if it satisfies the Cauchy criterion.

Activity

Corollary 14.5. If a series $\sum a_n$ converges, then $\lim a_n = 0$.

Prove this statement using Theorem 14.4.

Comparison Test

Thm. Let $\sum a_n$ be a series where $a_n \geq 0$ for all n. If $\sum a_n$ converges and $|b_n| \leq a_n$ for all n, then $\sum b_n$ converges.

Discussion.